

# Corporate Treasury and Liquidity Management Project at Unilever plc

*A public-data study of the trade-off between liquidity protection and the cost of holding cash*

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## 1. Executive summary

This report explains what was built, why it was built and what it is for. It documents a corporate treasury model of Unilever plc constructed entirely from public disclosures, and it is intended to accompany the Excel workbook and the Python analysis so that a reader can follow the reasoning without opening either.

The work proceeds in three layers. A historical model reconstructs the group's profitability, cash generation, debt and working capital between 2021 and 2025 from the annual reports. An Excel treasury model then projects cash week by week over thirteen weeks under three scenarios and measures the result against an explicit minimum liquidity buffer, deciding when funding is needed and when surplus cash can be invested. Finally, a Python layer reproduces the Excel model independently and, only once that reproduction is exact, uses it to examine how the size of the buffer should be chosen.

The question the model answers is a practical one: how much short-term liquidity should a large multinational hold, when idle cash earns less than the cost of borrowing, yet too little cash leaves it exposed if customers pay late or credit conditions tighten? Under the stated assumptions, the smallest buffer that absorbs the severe scenario's cash drain without recourse to the committed facilities, while keeping any borrowing within those facilities, is €3.5bn. The Excel model and the Python analysis arrive at this figure separately.

The €3.5bn figure is a model output under stated assumptions. It is not Unilever's liquidity policy, which the company does not disclose, and it should not be read as a recommendation to the company.

## 2. Objective and treasury question

The project asks how a multinational should manage short-term liquidity while balancing liquidity protection against funding and opportunity costs. The model uses Unilever's disclosed 2025 average short-term borrowing rate of 4.3% as a funding-cost proxy, compared with a 2.0% modelled short-term investment yield, so cash held in reserve carries a cost. Against that carrying cost sits the risk of a shortfall: a sudden need to draw on credit facilities at short notice, at a moment when lenders and investors are most attentive. The choice of buffer is therefore a choice between the cost of holding cash and the resilience it buys, and that choice depends heavily on how reliably inflows can be forecast.

The question lies at the centre of corporate treasury and of the transaction banking businesses that serve it. Thirteen-week cash forecasting, committed revolving credit facilities, cash pooling, supplier finance and short-term investment products are the working tools of both.

### Why Unilever?

Unilever was selected because its scale, international operations, geographically dispersed cash, supplier finance exposure and substantial committed facilities create a realistic corporate treasury problem, while its detailed public disclosures make the analysis reproducible. Around 85% of its cash is held in subsidiaries outside its central finance companies, €2.7bn of its trade payables sit within supplier finance programmes, and it maintains roughly €7.0bn of undrawn committed facilities.

### 3. Data and historical analysis

The source material comprises Unilever's annual reports for 2022 to 2025, its full-year results announcements, the machine-readable financial statements filed with the United States Securities and Exchange Commission, and the company's restatement of 2023 and 2024 excluding the Ice Cream business. These were assembled into a raw dataset in which each figure is marked as reported, derived or unavailable, and which was then frozen so that no later calculation could overwrite it.

Before the historical model was accepted, a reconciliation pass returned to the primary documents to find the figures that the internal checks required. That pass corrected two errors in the compiled data: net finance costs for 2023 were €486m on the original basis rather than €409m, and €408m on the continuing basis. It also identified a line missing from the bridge between operating profit and profit before tax, namely the monetary loss arising from hyperinflationary economies. With these corrections, all six core checks (profit before tax, net income, financial liabilities, free cash flow, net debt and the cash bridge) reconcile in every period tested.

Particular care was needed over 2023. Following the demerger of the Ice Cream business, completed in December 2025, Unilever restated 2023 and 2024 on a continuing basis. The model therefore holds 2023 twice, as reported originally and as restated, and compares like with like: 2023 original with 2022, and 2024 with 2023 continuing. Treating the two bases as a single series would have produced an apparent fall in revenue of some €8bn that is an accounting consequence of the demerger rather than a change in trading.

Several observations shape what follows. Reported revenue moved from €52.4bn in 2021 to €50.5bn in 2025; on the continuing basis it moved from €51.7bn in 2023 to €50.5bn, largely because of currency, while operating margin rose from 17.4% to 17.9%. Free cash flow on Unilever's own definition was €5.9bn in 2025, compared with €6.4bn in 2021 and €6.3bn in 2024. Reported net debt fell from €25.5bn to €23.1bn over the period, and at the end of 2025 cash and other current financial assets amounted to €5.1bn. Working capital is strongly negative: in 2025 receivables represented 35 days of revenue, inventory 55 days of cost of sales and payables 150 days, the last inflated by supplier finance. One further point proved instructive. A simple comparison of year-end balances suggests that working capital released €1.2bn of cash in 2025, whereas the cash flow statement reports €0.1bn. The difference arises from Ice Cream balances leaving the balance sheet at demerger, and it shows why movements in balance-sheet items cannot be read as cash.

### 4. The Excel treasury model

The treasury model sits on top of the frozen historical model in twelve further sheets. Every assumption lives on a single sheet, set out in Base, Downside and Severe columns and classified as disclosed, derived or a modelling assumption. A single selector drives one forecast engine, so there are no parallel copies of the model for each scenario.

#### Thirteen-week forecast

The forecast begins on 1 January 2026 from the balance-sheet cash of €3,941m at 31 December 2025, which is kept distinct from the €3,870m shown in the cash flow statement because the latter is stated net of overdrafts. Annual magnitudes derived from the 2025 continuing figures are converted into weekly flows: customer receipts arrive evenly, payroll and tax fall in the weeks containing a month end, and maturing debt is spread across the year and assumed to be refinanced. Payroll is proxied by the disclosed employee benefits expense of €6.6bn. The next dividend, payable on 10 April 2026, falls just outside the horizon.

#### Scenarios

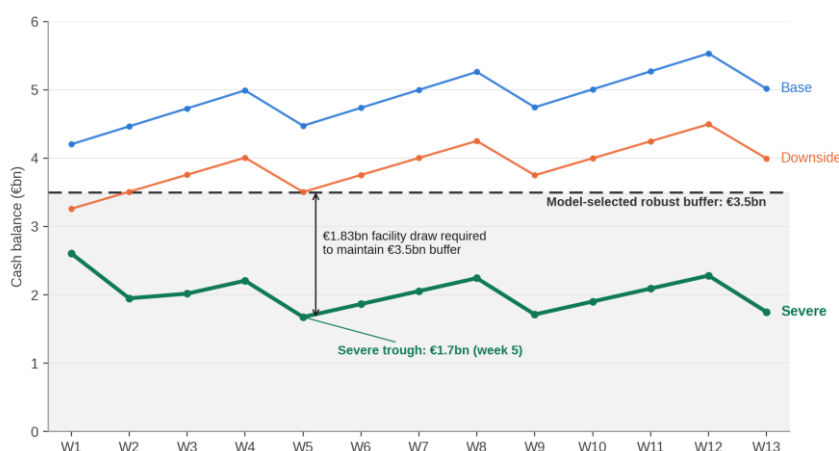
In the Downside case customers pay seven days later, floating rates rise by 50 basis points and currencies other than the euro fall by 5%. The Severe case combines collections 5% weaker and fifteen days later, five additional days of inventory, suppliers paid five days sooner, a 200 basis point rise in rates and a 10% currency move. The Downside delay already exceeds the largest annual change in receivable days observed in the data, an increase of 5.7 days between 2024 and 2025. No management action, such as cutting capital expenditure or suspending buybacks, is assumed in either case.

**Liquidity buffer, funding and surplus cash.** The modelled buffer covers three weeks of gross operating outflows, uplifted by 10% for stress, plus €160m of cash restricted by exchange controls, giving €3.0bn; periods of one to six weeks are tested alongside it. When cash would fall below the buffer, the model draws on the committed facilities of USD 5.2bn and EUR 2.6bn, about €7.0bn at the European Central Bank rate of 1.1750 at the year end, and repays as cash recovers. Drawings bear interest at the disclosed 2025 average short-term borrowing rate of 4.3% plus the scenario shock, while cash above the buffer is assumed to earn 2.0%, the ECB deposit facility rate. Both costs are reported beneath the cash line so that no circular reference arises between cash, borrowing and interest.

**Currency and controls.** A currency sheet asks whether exchange rate movements alter the thirteen-week cash requirement. It is kept deliberately modest and is distinguished from Unilever's €21.6bn net investment exposure, which is a balance-sheet translation exposure rather than a short-term cash one. Twelve core checks and four supplementary checks test the cash roll-forward, the reconciliation of every total, the funding and investment arithmetic and the integrity of the scenario selector; all pass in each scenario.

### 13-Week Liquidity Profile Under Stress

Severe conditions create a temporary liquidity shortfall despite positive underlying cash generation.



Cash balance = projected closing cash before liquidity funding, weeks beginning 1 Jan 2026 (opening balance-sheet cash €3.9bn). Shaded area lies below the robust buffer. Source: author's model built from Unilever public disclosures (annual reports 2022–2025, FY results, Form 20-F). Model output under stated assumptions; not Unilever's liquidity policy.

Figure 1. Weekly cash before liquidity funding under the three scenarios, against the €3.5bn robust buffer.

Figure 1 traces the three cases week by week. In the Base case cash never approaches the buffer. The Downside case leaves only €249m of headroom above the €3.0bn modelled buffer at its lowest point, and 3.7 further days of payment delay would require borrowing. In the Severe case cash falls to €1.7bn in the fifth week: €1.34bn must be drawn to restore the modelled buffer, equivalent to 19% of the committed facilities, and €1.83bn to maintain the €3.5bn robust buffer shown in the figure.

## 5. Python analysis

Python was used as a controlled analytical layer rather than as a second, competing model. It reads the frozen workbook, locating each figure by its row label, and never writes to it. Before any analysis it validates the inputs and then reproduces the Excel model independently from the imported assumptions.

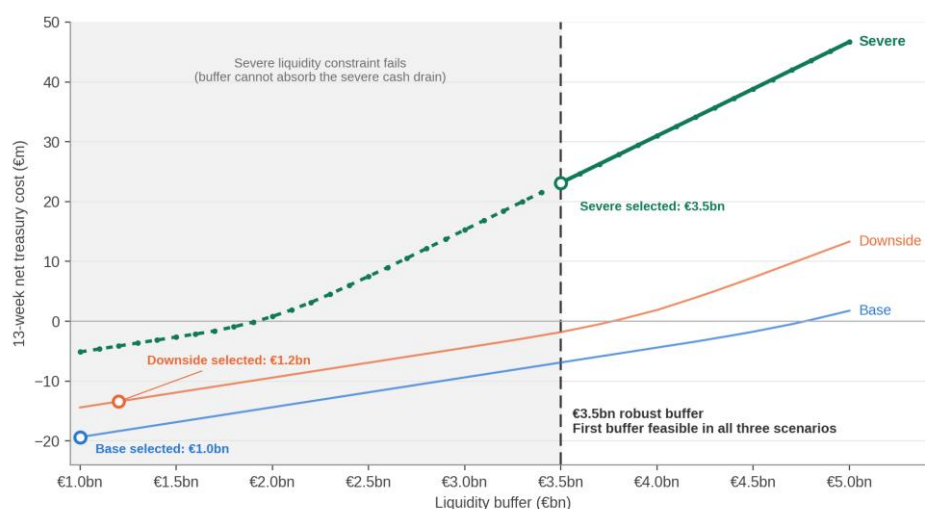
All 842 comparisons agree within a tolerance of €0.01m, the largest difference being of the order of  $10^{-11}$ . The comparisons cover every week of every scenario, every line of the Base forecast, the drivers, the headline measures, the optimisation table, both sensitivity grids and the reverse stress figures. Only after this test is passed does the analysis begin.

The buffer analysis evaluates levels from €1.0bn to €5.0bn in €100m steps for each scenario. A level is feasible if the buffer, net of restricted cash, absorbs the scenario's largest cash drain relative to the Base case, and if the borrowing needed to maintain it stays within the committed facilities. Among feasible levels the model selects the one with the lowest net treasury cost, defined as funding cost less investment income. A single robust buffer is then chosen as the level feasible in all three scenarios with the lowest worst-case cost. Sensitivity analysis varies the collection delay from nought to twenty days and the funding rate and investment yield by one percentage point either way, together with a two-way grid of delay against funding rate. A robustness exercise changes one assumption at a time, scaling the collection, working capital and currency stresses by a half and by one and a half.

Twenty-one economic tests guard the results; for instance, a longer collection delay must never improve the minimum cash balance, and a higher funding rate must never reduce funding cost. Two results that at first appeared counterintuitive were investigated rather than accepted. A larger buffer requires more borrowing to maintain, not less; what it reduces is the portion of stress the buffer cannot absorb. Likewise, a harsher currency shock slightly lowers the required buffer, because until the fifth week non-euro payments exceed non-euro receipts, so a stronger euro makes those net payments cheaper.

### Liquidity Buffer vs. 13-Week Treasury Cost

€3.5bn is the smallest buffer that satisfies the Severe liquidity constraint, not the cheapest buffer in isolation.



Net treasury cost = funding cost – investment income over 13 weeks (negative = net income). Grid search over buffers of €1.0bn–€5.0bn in €100m steps (dots = Severe grid points; dashed = infeasible). Feasible = buffer less restricted cash absorbs the scenario's peak cash drain vs Base, with borrowing inside committed facilities. Source: author's model built from Unilever public disclosures (annual reports 2022–2025, FY results, Form 20-F). Model output under stated assumptions; not Unilever's liquidity policy.

Figure 2. Liquidity buffer against thirteen-week net treasury cost across the €100m grid.

## 6. Results

### Scenario Summary

Each scenario's model-selected buffer, cash drain and peak facility requirement; the robust buffer covers all three.

|                                 | BASE   | DOWNSIDE | SEVERE  | ROBUST  |
|---------------------------------|--------|----------|---------|---------|
| Model-selected buffer           | €1.0bn | €1.2bn   | €3.5bn  | €3.5bn  |
| Cash drain vs Base              | —      | €1.04bn  | €3.27bn | €3.27bn |
| Peak funding to hold the buffer | €0     | €0       | €1.83bn | €1.83bn |
| Peak facility utilisation       | 0%     | 0%       | 26%     | 26%     |

### Robust buffer: €3.5bn

Model-selected result under stated assumptions; not Unilever's disclosed liquidity policy.

Funding and utilisation are measured at each column's own buffer (robust column: worst case across scenarios). Committed facilities €7.0bn (USD 5.2bn + EUR 2.6bn at 1.1750). At the €3.0bn modelled 3-week buffer, Severe would draw €1.34bn (19% of facilities). Source: author's model built from Unilever public disclosures (annual reports 2022–2025, FY results, Form 20-F).

Figure 3. Scenario summary at each scenario's model-selected buffer.

| Scenario                  | Model-selected buffer | Net treasury cost at that buffer (13 weeks) | What determines the selection                               |
|---------------------------|-----------------------|---------------------------------------------|-------------------------------------------------------------|
| Base                      | €1.0bn                | income of €19.3m                            | No constraint binds; lowest point of the grid               |
| Downside                  | €1.2bn                | income of €13.4m                            | Absorbs a €1.04bn cash drain relative to Base               |
| Severe                    | €3.5bn                | cost of €23.1m                              | Absorbs a €3.27bn cash drain relative to Base               |
| <b>Robust (all three)</b> | <b>€3.5bn</b>         | <b>worst case: cost of €23.1m</b>           | <b>Feasible in every scenario; matches the Excel result</b> |

Table 1. Model-selected buffers under stated assumptions (€100m grid).

**Collection timing is the dominant liquidity driver.** At the €3.0bn methodology buffer, peak borrowing in the Severe case rises from €44m with a four-day delay to €732m at ten days and €1.94bn at twenty, roughly €120m for each additional day, which is close to one day's customer receipts. Even the Base case would need to borrow once payments slip by more than about ten and a half days. In both the Downside and Severe cases, delayed collection accounts for the largest single share of the cash drain.

**Working capital stress matters.** Halving or increasing by half the inventory and supplier payment stresses moves the robust buffer between €3.1bn and €3.8bn; the equivalent change to the collection delay moves it between €2.6bn and €4.4bn. The two most sensitive inputs are therefore both working capital assumptions, which underlines why a treasury function depends on accurate receivables and payables data from the business.

## Liquidity Buffer Sensitivity

Collection timing has materially greater influence on buffer size than funding or investment rates.

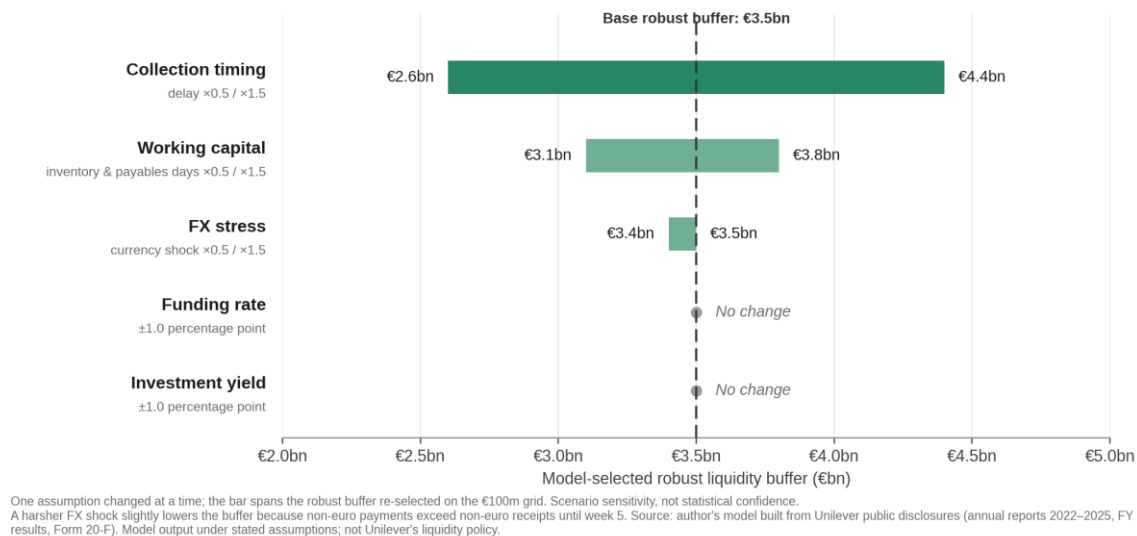


Figure 4. Range of the robust buffer when one assumption is changed at a time.

**The selection is driven by the constraint.** Net treasury cost rises with the size of the buffer in every scenario, so the cheapest feasible level is always the smallest one that meets the liquidity constraint. The €3.5bn result is the first level on the grid that covers the Severe drain of €3.27bn together with €160m of restricted cash. The economics of carry determine what the buffer costs; the stress determines how large it must be.

**Rates did not move the buffer.** Changing the funding rate or the investment yield by one percentage point left the robust buffer unchanged in every test. These rates alter the cost of the chosen buffer, with Severe net cost ranging from €13.0m to €17.9m across the funding rates tested, but they do not alter the liquidity constraint that fixes its size.

## 7. Assumptions and limitations

The model uses public data only. No confidential Unilever treasury information has been used or inferred, and the company's internal forecasts, cash pooling arrangements and policies are unknown to it. The weekly timing of receipts and payments is assumed, since no weekly data are published: receipts are spread evenly, payroll and tax fall in month-end weeks, and debt maturities are spread evenly across the year. The split of operating cash flows by currency is an illustrative assumption loosely informed by the geographic mix of the business, not disclosed data.

### Methodological limitations and future iterations

While the model establishes a source-reconciled and internally controlled baseline, it operates within three structural boundaries dictated by public-data availability.

### Capital Mobility vs. FX Price Shocks:

The FX module applies a linear shock to non-EUR operating flows. In a systemic crisis, liquidity risk may also arise from localised cash trapping, capital controls or restrictions on cross-border transfers, which could limit the effectiveness of central cash pooling. The model captures €160m of disclosed restricted cash but does not attempt to model these jurisdiction-level constraints.

**Funding Stack:**

For conservative backstopping, the engine routes liquidity deficits to the committed €7.0bn bank facilities. In live operations, an investment-grade corporate may have access to cheaper short-term Commercial Paper (CP) alongside bank facilities. CP is excluded because public disclosures do not provide sufficient information to model its availability, pricing or investor appetite under stress. Intra-Month Collection Seasonality: Due to public disclosure limits, weekly inflows are modelled using an even distribution. Real-world corporate collections can be lumpy around customer payment dates, meaning acute intra-week liquidity troughs may exist that an evenly phased model smooths over.

**Future iteration:**

With proprietary treasury data, these limitations could be addressed through jurisdiction-level cash-pooling constraints, a multi-layer funding hierarchy and daily rather than weekly cash-flow forecasting. These are also areas where transaction banks provide infrastructure and services to corporate treasurers.

The minimum liquidity buffer rests on a modelled methodology of three weeks of stressed outflows and is not Unilever's own measure. The 2.0% investment yield is a proxy based on the ECB deposit facility rate, and the 4.3% funding rate is anchored to the disclosed average short-term borrowing rate for 2025; neither represents the terms Unilever obtains from its banks. The Severe case is a stress scenario designed to test resilience; it is not a forecast of what will happen. The selection method is a search over a fixed grid rather than a continuous optimisation, and the robustness tests vary one assumption at a time, which is a form of sensitivity analysis rather than a statistical measure of confidence.

**Accordingly, the €3.5bn buffer is a model-selected result under the stated assumptions. It is not Unilever's actual liquidity policy and carries no recommendation.**

**8. Conclusion**

The project shows how a short-term liquidity decision can be analysed rigorously from public information. Its value lies less in the particular figure of €3.5bn than in the method that produces it: a source-reconciled historical base, a single transparent forecast engine with explicit assumptions and automated controls, and an independent reproduction in Python before any conclusion is drawn. The results make a clear point that is relevant well beyond Unilever. For a business of this kind the size of a prudent buffer is set chiefly by the uncertainty in collecting cash from customers and in managing working capital, while interest rates determine what that prudence costs. Better cash forecasting is therefore worth more than finer pricing, which is precisely the ground on which corporate treasurers and their transaction banks meet. The principal next step would be to replace the public-data proxies with daily, jurisdiction-level treasury data, particularly around cash mobility, funding access and collection timing.